UNIVERSITY OF PERPETUAL HELP SYSTEM

MOLINO CAMPUS

Name: Alexander Jon S. Solis

Section: BSCS-DS 3rd Year

Date: February 16, 2026

Answer the following questions and write your answer on the sheet provided below:

Performance Metrics Activity

Given:

- **Task A takes 50 seconds to complete.**
- **Task B takes 20 seconds to complete.**
- **The CPU is idle for 10 seconds in a 60-second observation.**
- **A system processes 200 requests in 5 minutes.**
- **After optimization, Task A takes 25 seconds.**

Questions:

1. **Compute the execution time for Task A and Task B.**

   The execution time of Task A is 50 seconds while the Task B's execution time is 20 seconds.

2. **Calculate throughput (requests per minute).**

   The throughput of the system is 40 requests per minute. Because 200requests/5minutes = 40 requests per minute

3. **Find the latency if each request starts 0.5 seconds after the previous one.**

   The latency is 0.5 seconds or 500ms.

4. **Compute CPU utilization.**

   The CPU utilization is at 83.33%. The solution is: (60 – 10) / 60 x 100 = 83.33%.

5. **Determine the speedup for Task A after optimization.**

   The task A increased its speed for 2x after the optimization. This is the formula used:

   Speedup = old execution time / new execution time

   Speedup = 50/25 = 2

6. **Discuss how scalability could improve if another CPU is added to handle more tasks.**

   Adding another CPU improves the scalability by allowing the computer to process multiple tasks in parallel. It can reduce waiting time, increases the throughput and balances the workload across the processors.